LESSON 5

Continuous probability distributions

UNIVERSITAT POLITÈCNICA DE VALÈNCIA

Aida Villalba Ortiz
aivilor@upv.es

Index

Aida Villalba Ortiz
aivilor@upv.es

Uniform distribution U(a,b) - Definition

The domain of the uniform distribution is defined by two parameters, a and **b**, which are its minimum and maximum values.

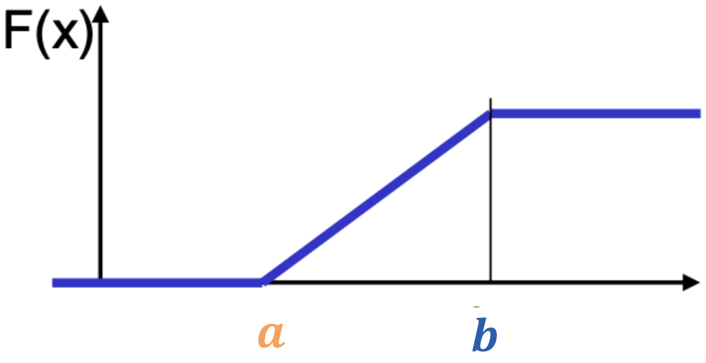
$\mathbf{U}(\boldsymbol{a}, \boldsymbol{b}) \sim \boldsymbol{X}$ where **a** is the lower limit and **b** is the upper limit

- The probability that **x** belongs to any subinterval S of the interval **[a,b]** is proportional to the length of the subinterval.
- The random variable has constant probability density in [a,b] and zero elsewhere.
- All intervals of equal length in the distribution in its range are equally probable.

Uniform distribution U(a,b) - Definition

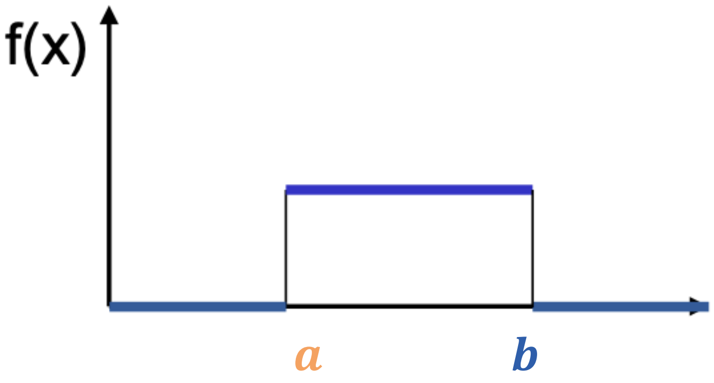
Distribution function

$$F(X) = \begin{cases} 0 & \forall\, x < a \\ \dfrac{x-a}{b-a}0 & \forall\, x \in [a,b] \\ 1 & \forall\, x > b \end{cases}$$



Density function

$$f(x) = \begin{cases} \dfrac{1}{b-a} & \forall\, x \in [a,b] \\ 0 & otherwise \end{cases}$$



Uniform distribution U(a,b) - Definition

Mean:

$$E(X) = \frac{a+b}{2}$$

Variance:

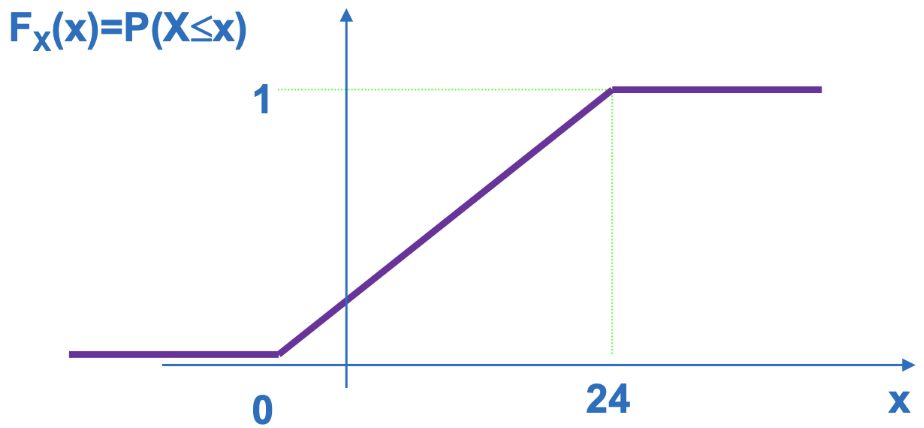
$$\sigma^2(X) = \frac{(b-a)^2}{12}$$

Standard deviation:

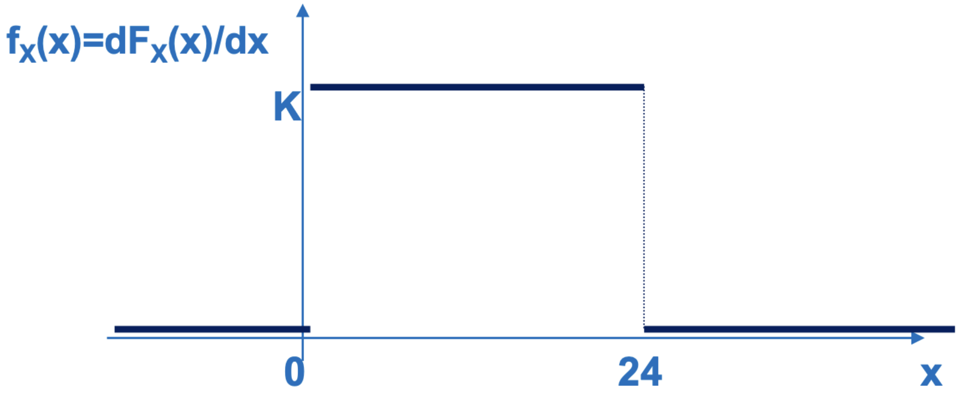
$$\sigma(X) = \sqrt{\frac{(b-a)^2}{12}}$$

Uniform distribution U(a,b) - Example

X: time of machine failure $x \in [0,1,\ldots,24]$



$$F_X(x)=\begin{cases} 0 & x \leq 0 \\ x/24 & x\in(0,24) \\ 1 & x \geq 24 \end{cases}$$



$$f_X(x)=\begin{cases} 0 & x \leq 0 \\ 1/24 & x \in (0,24) \\ 0 & x \geq 24 \end{cases}$$

Exercise 1

A train departs from a station every 20 minutes. A traveler arrives unexpectedly and takes the first train leaving the station. Find:

a) Distribution function of the random variable “Traveler's waiting time”.

b) Probability that the traveler will wait less than 7 minutes for the train.

c) The probability that he will wait for the train between 4 and 11 minutes.

d) The average waiting time for the traveler.

e) The probability that he will wait exactly 12 minutes.

f) The probability that he will have to wait at least 5 more minutes if he has already been waiting for 6 minutes

Exponential distribution Exp(α) - Definition

It is used as a model when it is assumed that there is no product wear (ageing or material fatigue) and that, therefore, failures are accidental.

Exp$(\alpha) \sim T$ where α is the failure rate or failure per unit.

- The random variable indicates the lifetime of a product. The time **t** that elapses before an accidental failure occurs in its operation.
- Models the random duration of components over their useful life.

Exponential distribution Exp(α) - Definition

Distribution function

$$F(X) = P(X \leq x) = \int_{0}^{x} f(x)dx = 1 - e^{-\alpha x}$$

Density function

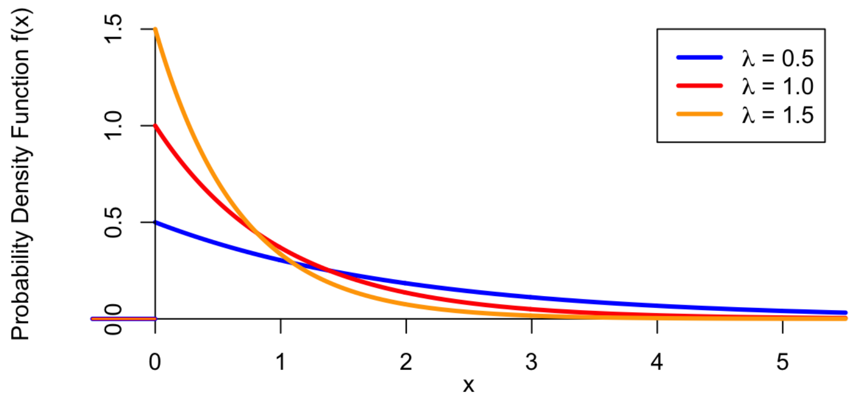
$$f(x) = \alpha \cdot e^{-\alpha x}$$

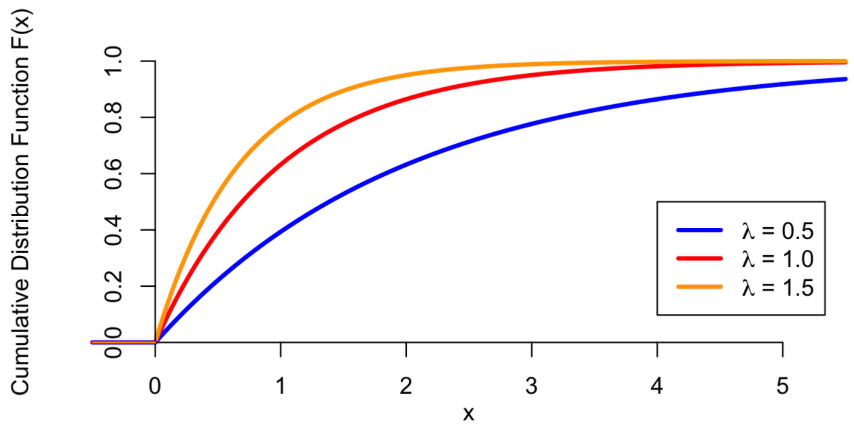
where $x \geq 0$ and $\alpha \geq 0$

Exponential distribution Exp(α) - Definition

Examples of usage

- Time until failure
- Time taken to complete a process
- Life of people, animals or physical components
- Duration of strikes
- Period of unemployment





Exp(α) – Mathematical expectation

Mean:

$$E(X) = \int_0^{\infty} f(x)dx = \alpha^{-1} = \frac{1}{\alpha}$$

Variance:

$$\sigma^2(X) = \int_0^{\infty} (x-m)^2 \cdot f(x)dx = \alpha^{-2} = \frac{1}{\alpha^2}$$

Standard deviation:

$$\sigma(X) = \sqrt{\int_0^{\infty} (x-m)^2 \cdot f(x)dx = \alpha^{-2}} = \sqrt{\frac{1}{\alpha^{-2}}}$$

Exp(α) – Invariability over time

Future behaviour is independent of your current or past behaviour:

$$P(X > t_0 + x / X > t_0) = P(X > x)$$

Failures are due to accidental causes and not to wear and tear (ageing) of the product. If failures in a machine become **more frequent as it ages**, the **time to failure does not follow an exponential distribution**.

Exp(α) – Exercise 2

It has been verified that the lifespan of certain elements follows an exponential distribution with a mean of 8 months.

a) Calculate the probability that an item will exceed 8 months of operation.

b) If the item has been operating for 8 months, what is the probability that it will end up operating for more than 16 months?

c) What conclusion can be drawn from the results in sections a) and b)?

d) Calculate the probability that an item will have a lifespan of between 3 and 12 months.

e) The 0.95 percentile of the distribution.

f) Calculate the median lifespan of these items.

g) Can this be considered a symmetrical variable?

Exp(α) – Exercise 3

The lifetime (until accidental failure) of electric batteries from a certain manufacturer is distributed exponentially with a mean of 50 hours.

a) What is the probability that a battery will fail within the next 10 hours?

b) If a battery has already been operating without accidental failure for 50 hours, what is the probability that it will fail accidentally within the next 10 hours?

c) What is the probability that a battery will last between 20 and 40 hours without accidental failure?

d) What is the length of time until accidental failure exceeded by 50% of the batteries?

Exp(α) – Exercise 3

Before calculating, consider whether or not it will be greater than the average life. If the manufacturer produces a new type of 'Extra Resistant' battery and informs us that 30% of the batteries exceed 300 hours before accidental failure

e) What is the probability that an extra-resistant battery will exceed 100 hours before accidental failure?

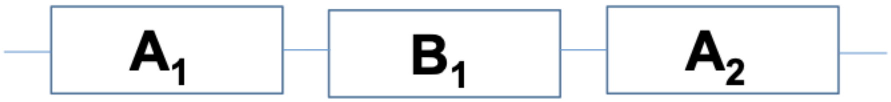
Exp(α) – Exercise 4

There are two types of components from a supplier (type A and type B) whose lifespans are distributed exponentially and independently.

a) 75% of type A components exceed 1,000 hours of operation. What is the probability that a type A component randomly selected from a batch from this supplier will exceed 500 hours?

b) Type B components have a median lifetime of 750 hours. What is the probability that a type B component randomly selected from a batch from this supplier will exceed 500 hours?

c) If we assemble a device like the one in the figure, what is the probability that the device will exceed 500 hours of operation?



Normal distribution N(μ, σ) - Definition

The normal distribution is the most important probability distribution in statistics, due to the large number of variables that are distributed according to it.

$$\mathbf{N}(\boldsymbol{\mu}, \boldsymbol{\sigma}) \sim \boldsymbol{X}$$

X follows a normal distribution, with mean μ and standard deviation $\boldsymbol{\sigma}$, with:

$$-\infty < \boldsymbol{X} < +\infty$$

and

$$\boldsymbol{\sigma} > \mathbf{0}$$

Normal distribution N(μ, σ) - Definition

Distribution function

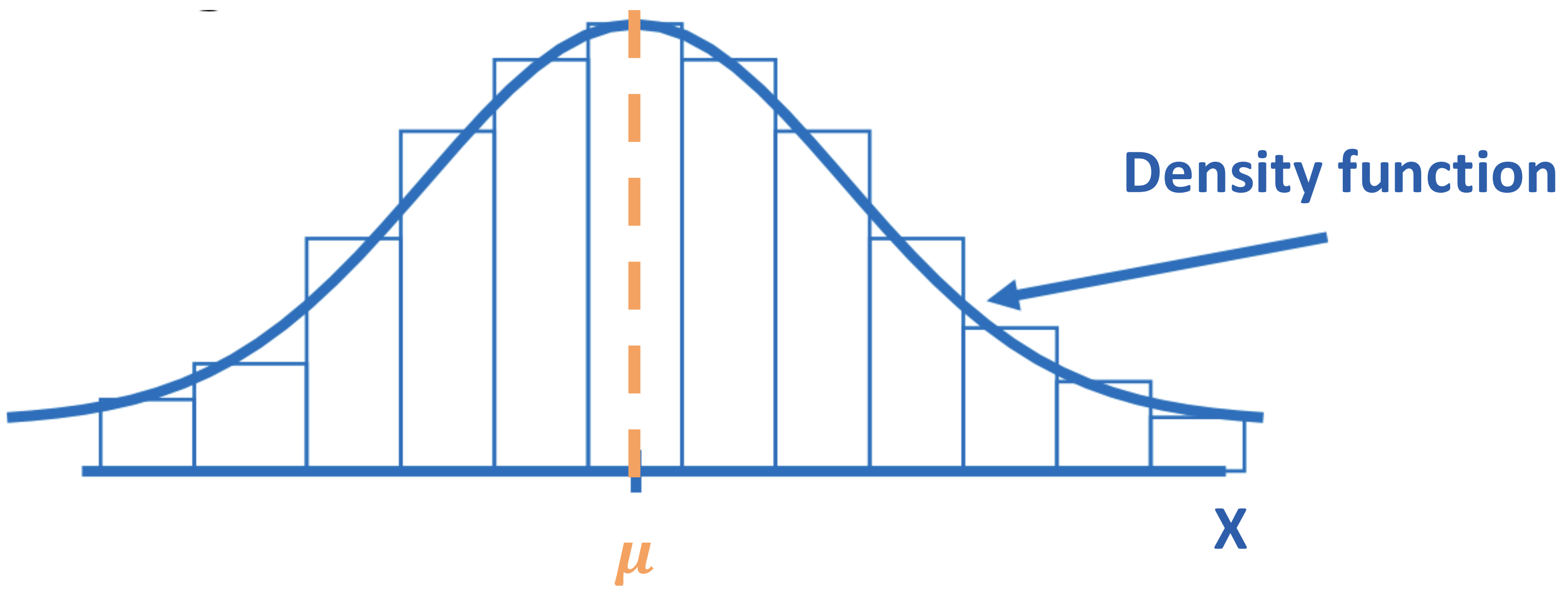
$$F(X) = P(X \leq x) = \int_{-\infty}^{x} f(x)dx = \textit{there is not primitive notion}$$

Density function

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

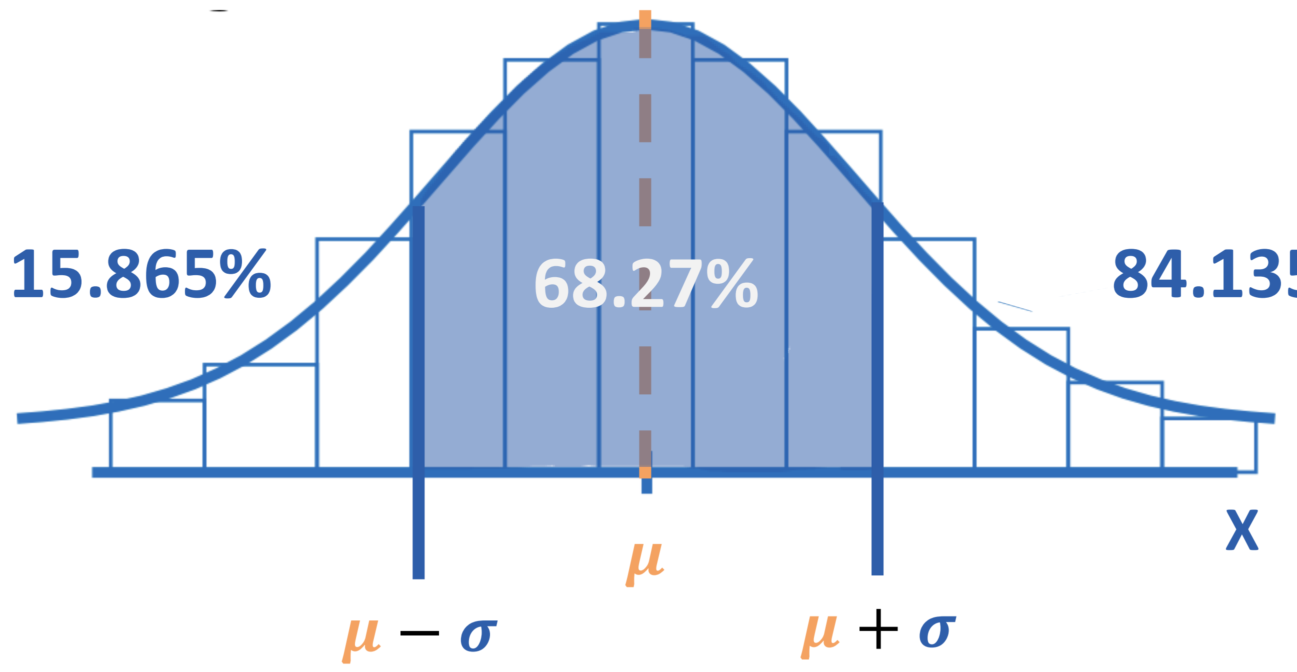
Normal distribution N(μ, σ) - Definition

Histogram of a normal variable X



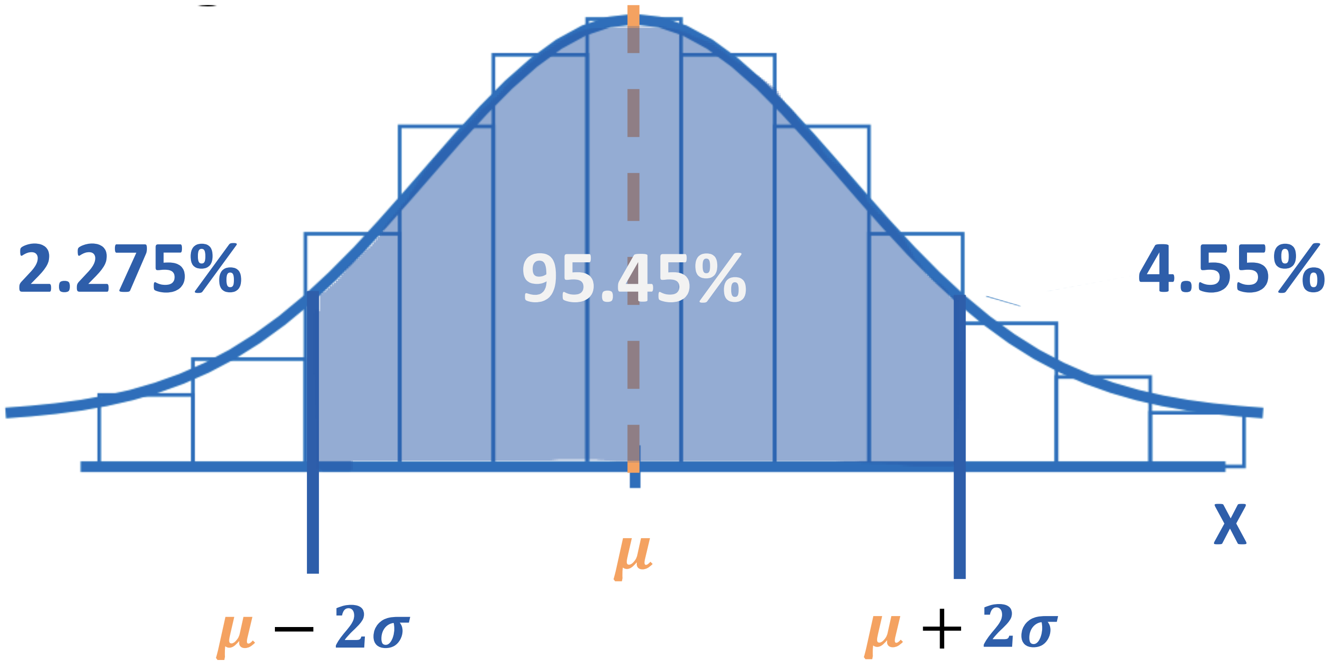
Normal distribution N(μ, σ) - Definition

$$P(\mu - \sigma < X < \mu + \sigma) = 0.6827$$



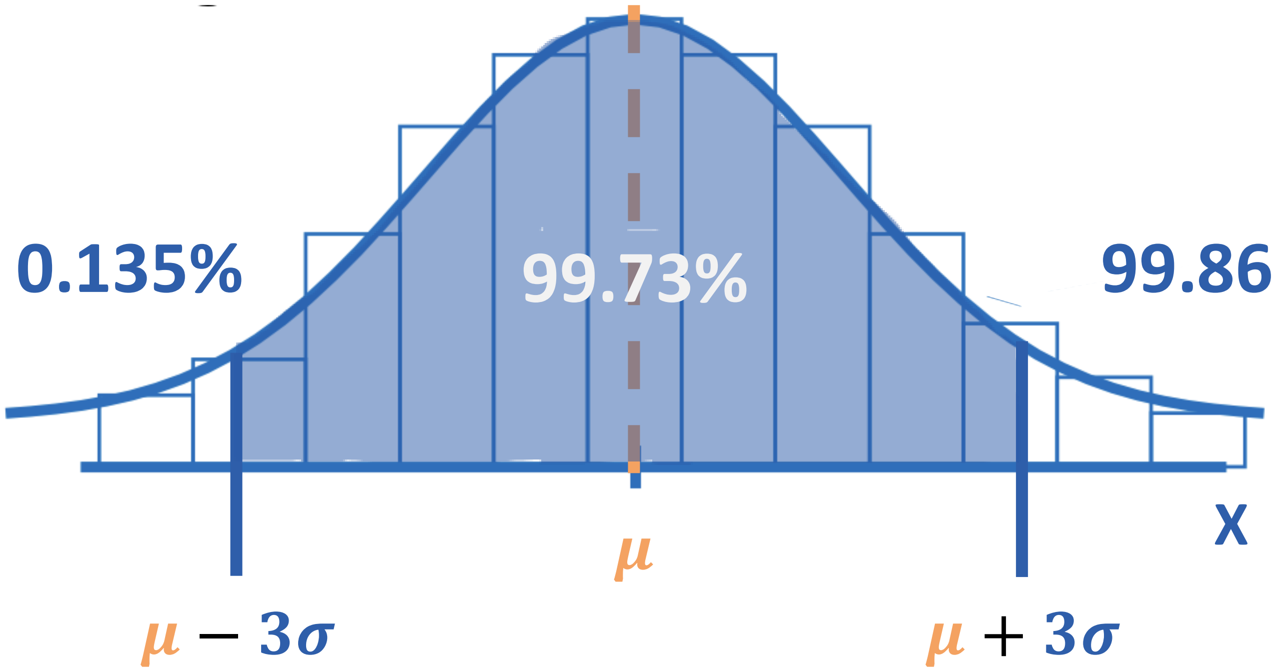
Normal distribution N(μ, σ) - Definition

$$P(\mu - 2\sigma < X < \mu + 2\sigma) = 0.9545$$



Normal distribution N(μ, σ) - Definition

$$P(\mu - 3\sigma < X < \mu + 3\sigma) = 0.9973$$



Normal distribution N(μ, σ) - Definition

Properties:

If $X \sim N(\mu_x, \sigma_x)$:

$$Y = a + bX \sim N\left(\mu_y = a + b\mu_x\,, \sigma_y = b\sigma_y\right)$$

If $X \sim N(\mu_x, \sigma_x)$ and $Y \sim N\left(\mu_y, \sigma_y\right)$ are **independent**:

$$Z = X \pm Y \sim N(\mu_z = \mu_x \pm \mu_y\,, {\sigma_z}^{\mathbf{2}} = {\sigma_x}^{\mathbf{2}} + {\sigma_y}^{\mathbf{2}})$$

N(μ, σ) - Mathematical expectation

Mean:

$$E(X) = \boldsymbol{\mu}$$

Variance:

$$\sigma^2(X) = \sigma^2$$

Standard deviation:

$$\sigma(X) = \sigma$$

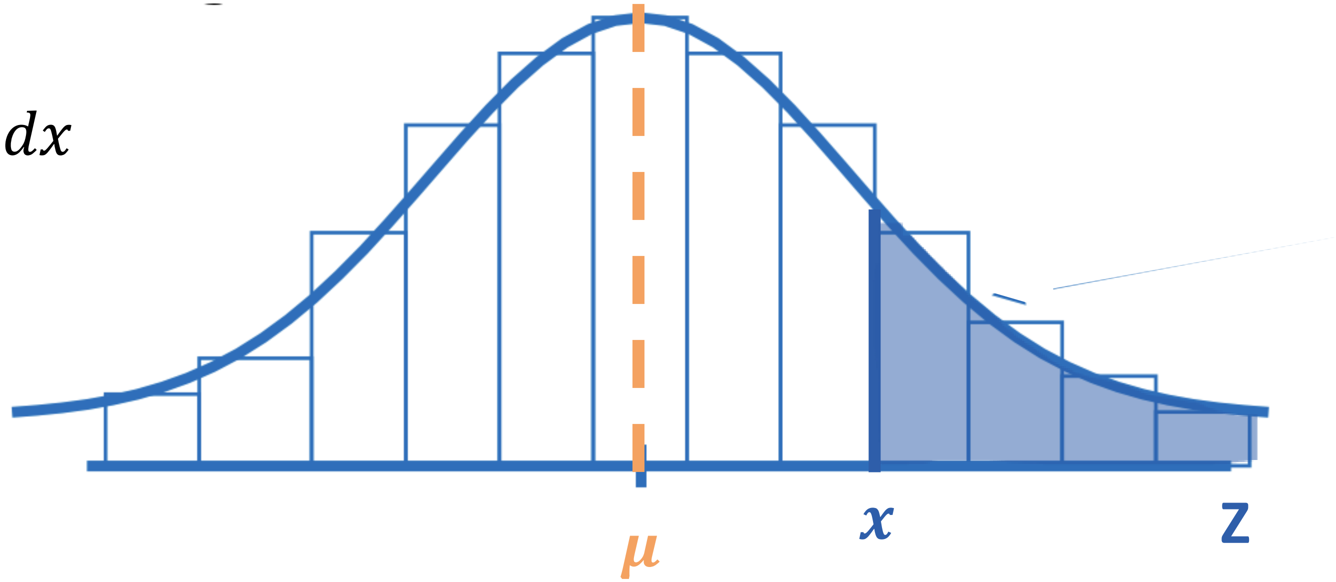
$N(\mu, \sigma)$ – Normal standardization

How to calculate $P(X \sim N(\mu, \sigma) > x)$?

$$P(X > x) = \int_{x}^{+\infty} \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx$$

↓

Analytically unsolvable

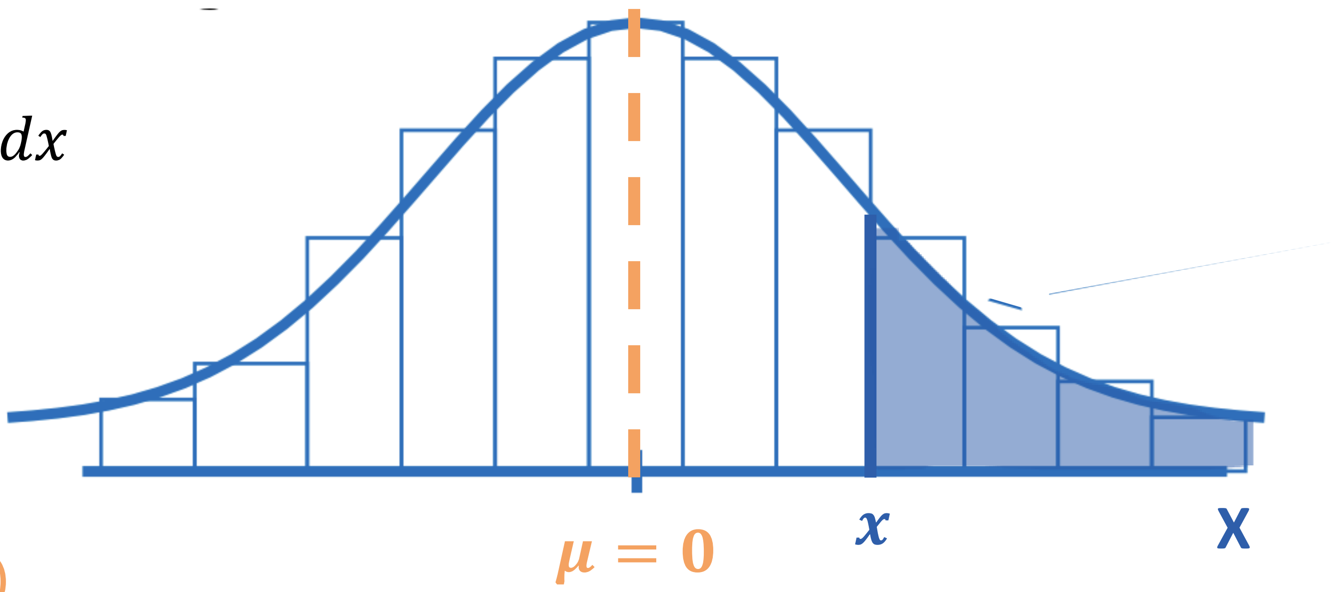


N(μ, σ) – Normal standardization

How to calculate $P(X \sim N(\mu, \sigma) > x)$?

$$P(X > x) = \int_{x}^{+\infty} \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx$$

$$Z = \frac{X - \mu_x}{\sigma_x} \sim N(\mu_z = 0 \, , \sigma_z = 1)$$



N(μ, σ) – Normal standardization table

z	0	1	2	3	4	5	6	7	8	9
-3,	*0,0013*	*0,0010*	*0,0007*	*0,0005*	*0,0003*	*0,0002*	*0,0002*	*0,0001*	*0,0001*	*0,0000*
-2,9	0,0019	0,0018	0,0018	0,0017	0,0016	0,0016	0,0015	0,0015	0,0014	0,0014
-2,8	0,0026	0,0025	0,0024	0,0023	0,0023	0,0022	0,0021	0,0021	0,0020	0,0019
-2,7	0,0035	0,0034	0,0033	0,0032	0,0031	0,0030	0,0029	0,0028	0,0027	0,0026
-2,6	0,0047	0,0045	0,0044	0,0043	0,0041	0,0040	0,0039	0,0038	0,0037	0,0036
-2,5	0,0062	0,0060	0,0059	0,0057	0,0055	0,0054	0,0052	0,0051	0,0049	0,0048
-2,4	0,0082	0,0080	0,0078	0,0075	0,0073	0,0071	0,0069	0,0068	0,0066	0,0064
-2,3	0,0107	0,0104	0,0102	0,0099	0,0096	0,0094	0,0091	0,0089	0,0087	0,0084
-2,2	0,0139	0,0136	0,0132	0,0129	0,0125	0,0122	0,0119	0,0116	0,0113	0,0110
-2,1	0,0179	0,0174	0,0170	0,0166	0,0162	0,0158	0,0154	0,0150	0,0146	0,0143
-2,0	0,0228	0,0222	0,0217	0,0212	0,0207	0,0202	0,0197	0,0192	0,0188	0,0183

$$Z = \frac{X - \mu_x}{\sigma_x} \sim N(\mu_z = 0\,, \sigma_z = 1)$$

If X ~ $N(0,1) \rightarrow P(X \leq -2.25) = 0.0122$

N(μ, σ) – Central Limit Theorem

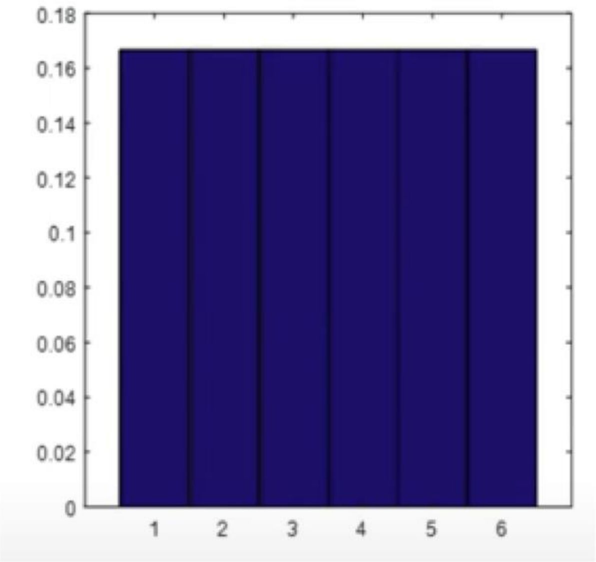
In very general terms, the sum of independent random variables tends to be normally distributed as the number of addends increases.

This theorem justifies the fact that **most distributions** of variables in **real problems** are normal.

$$\boldsymbol{Y = X_1 + X_2 + \cdots + X_N \xrightarrow[N \to \infty]{} Y \approx N(\mu, \sigma)}$$

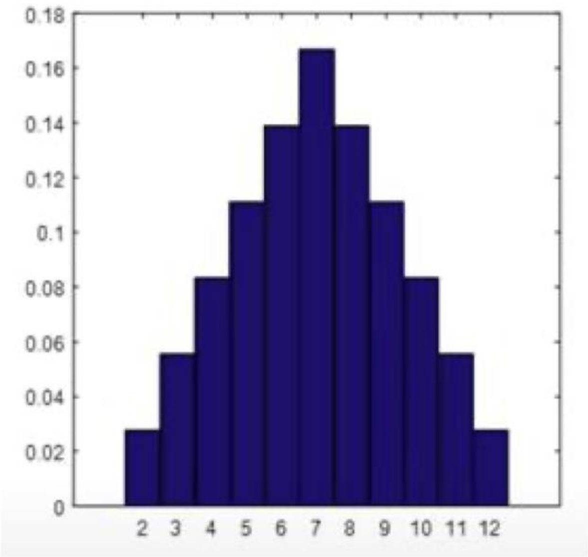
N(μ, σ) – Central Limit Theorem

Probability function sum value 1 given



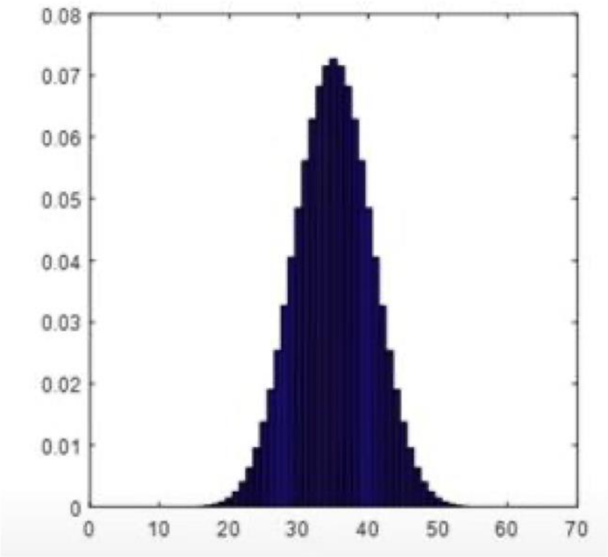
$$X = X_1$$

Probability function sum value 2 given



$$X = X_1 + X_2$$

Probability function sum value 10 given



$$X = X_1 + X_2 + X_3 + \\ + X_4 + X_5 + X_6 + X_7 + \\ + X_8 + X_9 + X_{10}$$

N(μ, σ) – Exercise 5

The hardness of parts manufactured in a factory normally fluctuates with an average of 185 Nw and a standard deviation of 12 Nw. What percentage of the parts manufactured will meet the established specifications of 180$\pm$20 Nw?

N(μ, σ) – Exercise 6

The diameter of the oranges in each batch arriving at a packing warehouse is normally distributed. It is known that 10% of a batch has a diameter of less than 40 mm and that 85% does not exceed 80 mm. Calculate the percentage of the batch whose diameter is between 60 and 70 mm.

N(μ, σ) – Exercise 7

X: The average daily temperature in Valencia

It is distributed as a normal distribution with a mean of 18ºC and a standard deviation of 1.5ºC.

a) What is the third quartile? What does this value mean?
b) What is the probability that this value will be exceeded at some point in a 10-year sequence? (assume independence)

N(μ, σ) – Exercise 8

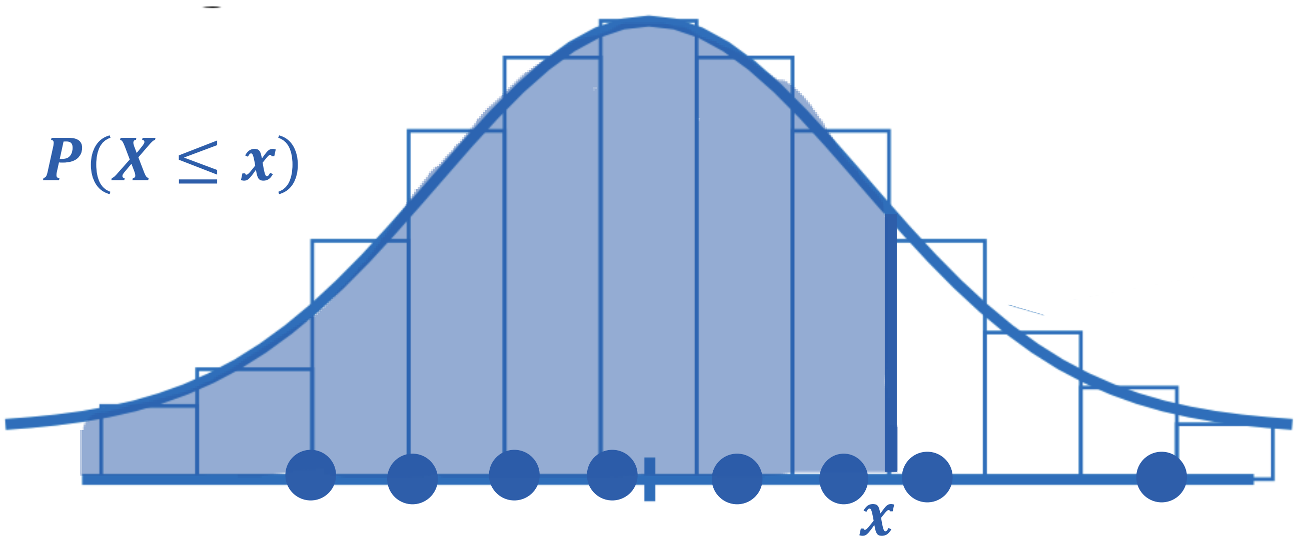
The height of the boys in the statistics class is normally distributed with a mean of 175 cm and a standard deviation of 10 cm, while the height of the girls is normally distributed with a mean of 164 cm and a standard deviation of 7 cm.

a) Calculate the probability that, when choosing a couple at random, the height of the girl will be greater than that of the boy.

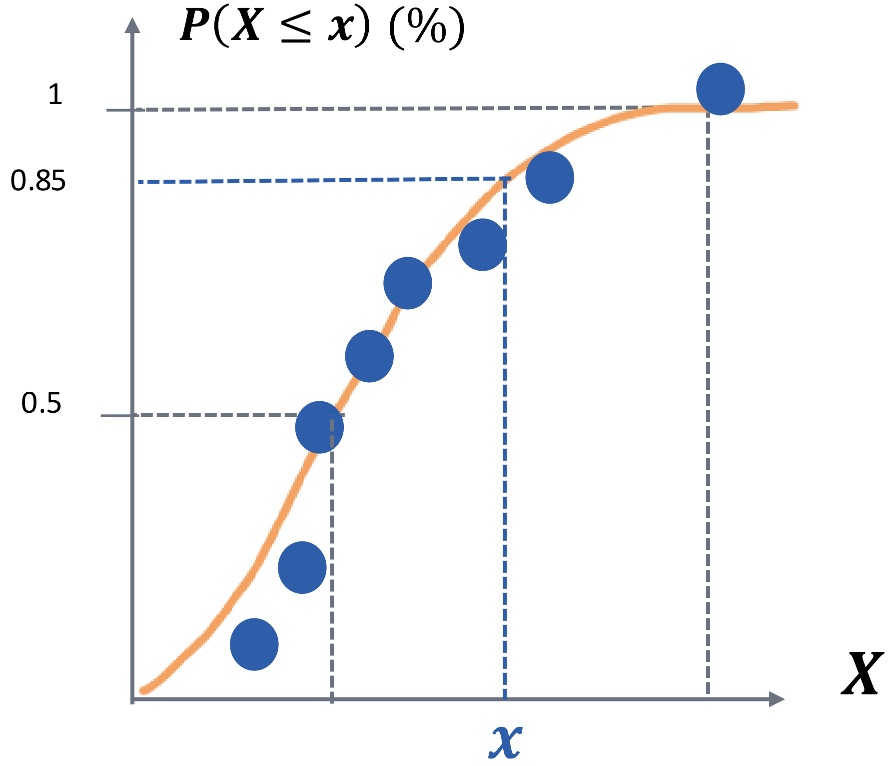
b) What is the probability that the taller member of the couple will be over 180 cm tall?

Normal Probability Plot

Normal density distribution plot

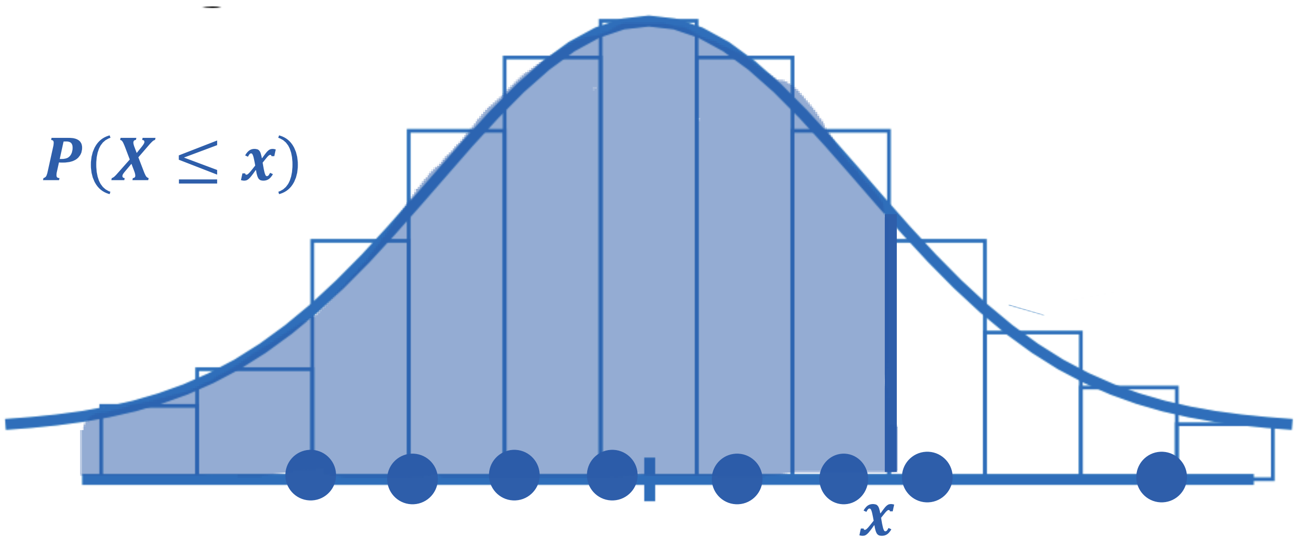


Plain graph paper



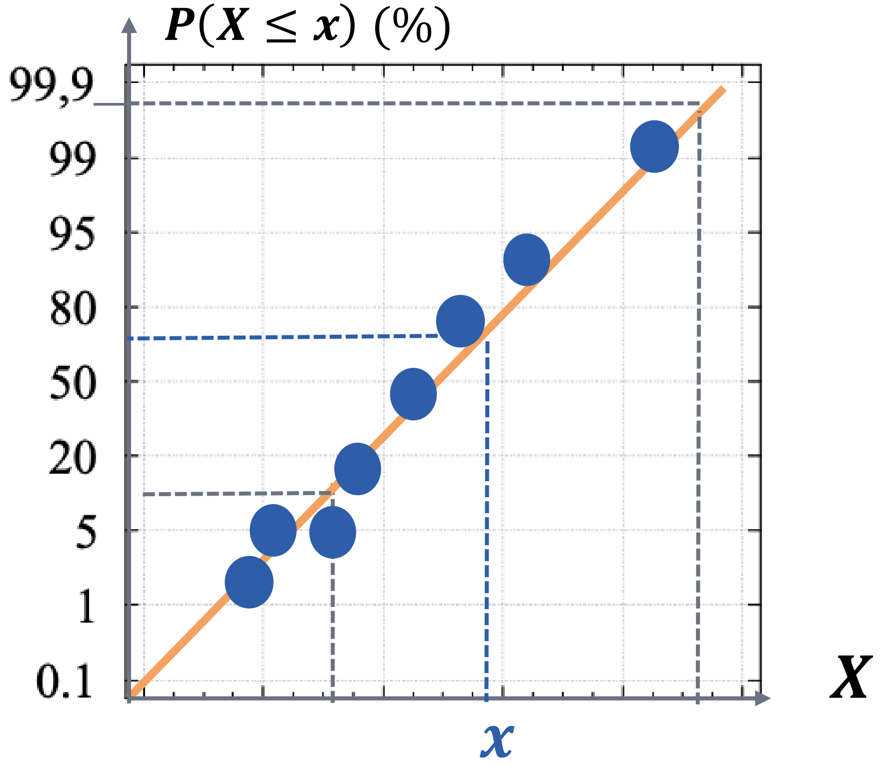
Normal Probability Plot

Normal density distribution plot



$$P(X \leq x)(\%) = \frac{i - \frac{1}{2}}{n} \cdot 100$$

Normal Probability Plot



Normal Probability Plot

HOW TO BUILD IT

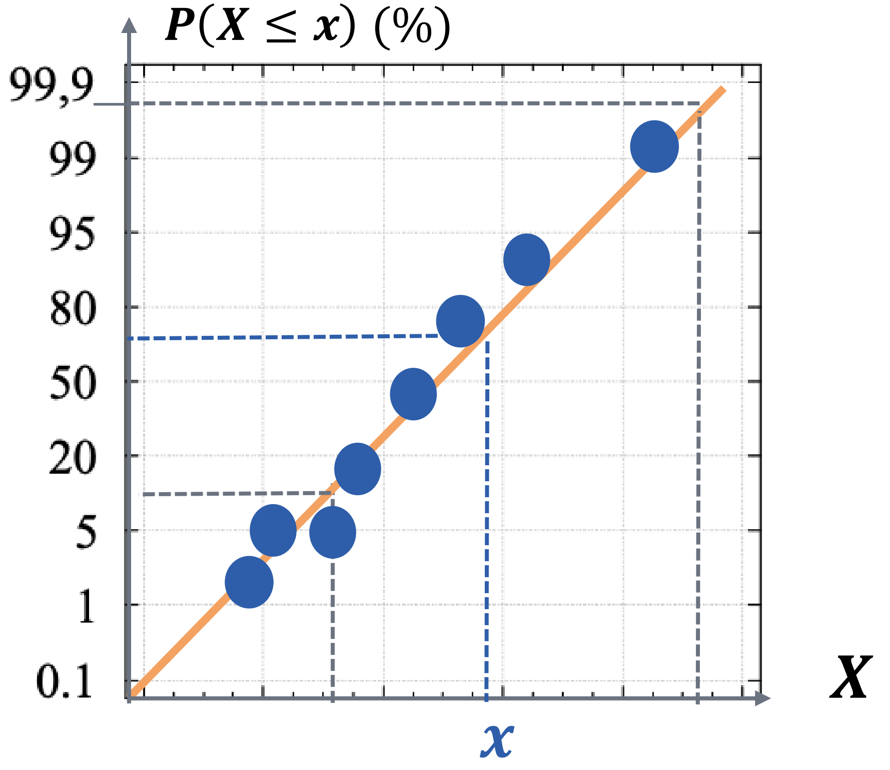
1) The data is sorted from lowest to highest because it represents cumulative relative frequencies and, consequently, the relative position of the data is considered rather than its actual value.

2) For each data value, which will be represented on the x-axis, its corresponding ordinate value is calculated using the expression:

$$P(X \leq x)(\%) = \frac{i - \frac{1}{2}}{n} \cdot 100$$

where 'i' is the position of the data and 'n' is the number of data points in the sample.

Normal Probability Plot



Normal Probability Plot - Exercise 9

Temperature (ºC) for 10 patients

36.9 37.2. 36.7 37.2. 37 36.9. 37.4. 37.6. 37.7. 37.5

a) Draw a normal probability graph for the sample above.

b) Estimate the median.

c) Estimate the mean and standard deviation.

d) Draw the box and whisker plot.

Normal Approximation

Binomial to Normal:

When $n \cdot p \cdot (1 - p) \geq 9$:

$$\boldsymbol{Bi(n,p)} \xrightarrow[\boldsymbol{n \cdot p \cdot (1-p) \geq 9}]{} \boldsymbol{N(\mu = n \cdot p, \sigma = \sqrt{n \cdot p \cdot (1 - p)})}$$

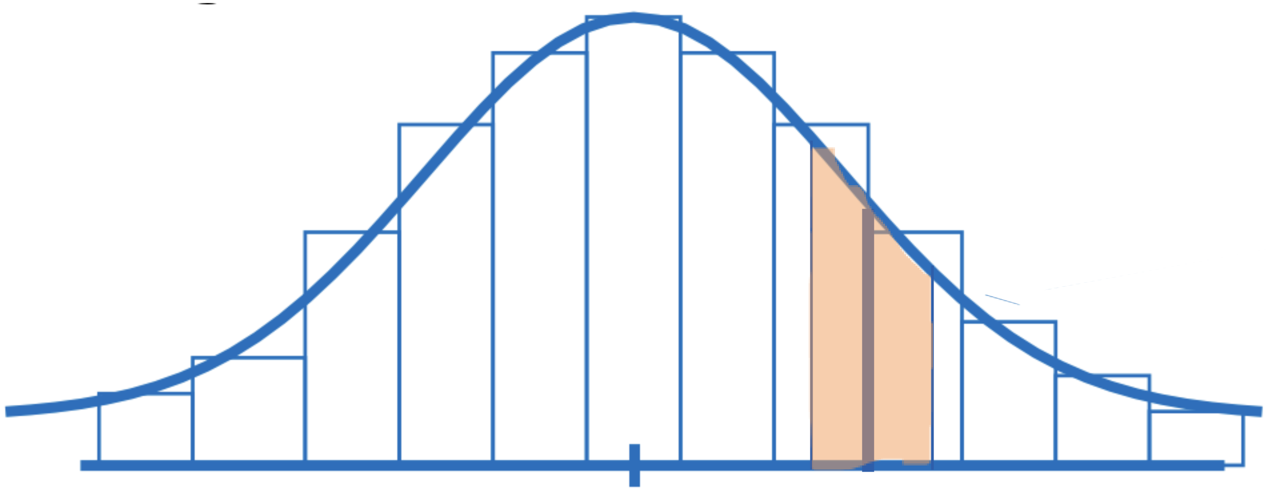
Poisson to Normal:

When $\lambda \geq 9$:

$$\boldsymbol{Ps(\lambda)} \xrightarrow[\boldsymbol{\lambda \geq 9}]{} \boldsymbol{N(\mu = \lambda, \sigma = \sqrt{\lambda})}$$

Continuity correction

When using the normal model to calculate the probabilities associated with a **discrete random variable**, as a result of approximation, a continuity correction must first be made.



$$P(X = x) = P(x - 0.5 \leq X \leq x + 0.5)$$

DISCRETE		CONTINUOUS
$P(X > 25)$	→	$P(X' \geq 25.5)$
$P(X \geq 25)$	→	$P(X' \geq 24.5)$
$P(X < 24)$	→	$P(X' \leq 23.5)$
$P(X \leq 24)$	→	$P(X' \leq 24.5)$

Exercise 10

A telephone switchboard receives an average of 2 calls per minute. Calculate approximately the probability that more than 150 calls will be received in one hour.

Exercise 11

Assuming that the probability of saying an odd digit is the same as that of an even digit, approximately calculate the probability that out of 131 randomly said digits, 88 or more will be odd.

Exercise 12

A company that sells chocolates sells boxes of 20 chocolates. If the weight of each chocolate follows a normal distribution of m=10 g and σ=1 g, the weight of the box containing them follows a normal distribution of m=50 g and σ=10 g.

a) What is the percentage of boxes of chocolates sold whose total weight exceeds 250 g?

b) What is the weight that will exceed 20% of the boxes of chocolates sold?

c) Find a weight range in which the central 80% of the boxes of chocolates sold will fall.

d) If I weigh all the chocolates in a box, what is the probability that there will be more than one chocolate weighing more than 12 grams?

Exercise 12

The company sells another type of special chocolate box (type E) whose total weight follows a normal distribution and which in 30% of cases exceeds 550g and in 60% of cases exceeds 530g.

e) What is the average total weight of type E boxes of chocolates?

f) What percentage of type E boxes of chocolates will weigh between 520g and 540g?

If the company sends an order of 200 boxes of type E chocolates and there is a 5% probability that these boxes will be underweight

g) What is the probability that there will be more than 13 underweight boxes in this shipment?